

SKILL LINK

Skill Link is an AI-driven digital platform designed to connect skilled workers with customers who require reliable household, technical, or professional services. The system integrates intelligent algorithms for personalized worker recommendations, sentiment-based feedback evaluation, geolocation-guided matching, and dynamic price prediction. It incorporates face-recognition authentication and verified user registration to strengthen security and build trust among service providers and clients. Skill Link is developed using PHP and MySQL and enhanced with machine learning modules that automate the selection and allocation of workers. By improving visibility for semi-skilled workers and simplifying service discovery for customers, the platform aims to promote equitable employment opportunities and digital participation within local communities.

Keywords— AI, Machine Learning, Service Recommendation, Geolocation, Chatbot System, Sentiment Analysis, Price Prediction, Face Recognition, Employment Platform, PHP, MySQL.

1. INTRODUCTION

In many developing nations, individuals with technical or manual skills face difficulty securing stable employment due to limited digital exposure and lack of organized platforms. Customers, on the other hand, struggle to identify trustworthy workers for everyday services such as plumbing, electrical repairs, beauty services, carpentry, and household maintenance. This results in a persistent gap between available skills and accessible job opportunities.

Skill Link aims to close this gap by offering a unified digital platform where workers can showcase their expertise and customers can easily book verified professionals. The platform uses artificial intelligence to recommend suitable workers based on skills, distance, availability, and previous service quality. Features such as chatbot-based assistance, geolocation matching, price estimation, and face-recognition login contribute to an efficient, secure, and user-friendly ecosystem. By enabling intelligent automation and trustworthy worker identification, Skill Link ensures faster job allocation while promoting digital empowerment for skilled and semi-skilled individuals.

2. LITERATURE REVIEW

Recent research in AI-driven employment and service platforms highlights a strong shift toward intelligent automation for skill-task matching. Studies on hybrid recommendation systems show that combining collaborative filtering, content-based filtering, and clustering leads to more accurate worker allocation compared to traditional manual matching. Researchers have demonstrated that incorporating behavioural patterns, skill metadata, and contextual features significantly improves recommendation quality. Additionally, various works emphasize the importance of analyzing similarity between service requests and worker profiles using cosine similarity and feature vectors, which enhances the personalization capabilities of job-matching platforms. These findings establish a strong foundation for systems like Skill Link, which depend on multi-criteria filtering to identify the most suitable service providers for users.

Geolocation-based recommendation frameworks have also gained significant attention in recent studies, particularly for location-sensitive applications such as home services and on-demand labor marketplaces. Research suggests that integrating spatial data and distance calculations improves the efficiency of service delivery by prioritizing workers located closest to the customer. At the same time, natural language processing techniques have been effectively used to interpret textual feedback, enabling platforms to classify user reviews into positive, negative, or neutral sentiments. Such sentiment-driven decision support mechanisms help refine service rankings and enhance trust among users. Furthermore, chatbot technologies powered by NLP and NLU have proven useful in handling booking queries, user guidance, and conversational automation, thereby reducing the burden on human administrators.

In addition, predictive modeling studies highlight the usefulness of regression algorithms for estimating fair service pricing based on factors such as worker experience, distance, urgency, and demand fluctuations. Machine learning-based cost estimation systems have shown high reliability in maintaining transparency for both customers and workers. Research on biometric authentication, especially face-recognition techniques using Local Binary Pattern Histograms (LBPH), confirms strong accuracy and robustness for verifying user identity in digital platforms. Scholars also emphasize the broader social impact of AI-enabled employment ecosystems, noting that such platforms enhance digital inclusion for semi-skilled workers by improving access to job opportunities and increasing economic stability. Together, these contributions support the development of the Skill Link platform, which integrates recommendation logic, sentiment analysis, geolocation mapping, price prediction, and biometric verification to create a trustworthy and intelligent service-matching environment.

Table 2.1: Literature Summary

Ref. No.	Paper Title / Author(s)	Year	Summary / Drawbacks
[1]	AI-Base Recommendation System — Ramesh et al.	2023	Proposed a hybrid job-matching model combining collaborative and content-based filtering. Improved matching accuracy, but struggled with cold-start problems for new users.
[2]	ML-Based Skill Matching in Digital Platforms — Mehta & Sinha	2022	Introduced skill similarity and cosine-based recommendation for professional matching. Effective for structured data but less accurate for unverified worker profiles.
[3]	Sentiment Analysis for Service Feedback — Gupta et al.	2024	Used NLP techniques to classify user reviews and improve service quality insights. Lacked contextual understanding for mixed or sarcastic reviews.
[4]	Geolocation Recommendation Framework — Narayan & Thomas	2021	Developed a location-aware employment system for connecting nearby workers. Faced challenges in rural areas due to inconsistent GPS accuracy.
[5]	Chatbot Automation in Customer Portals — Dey & Banerjee	2023	Created an NLU-based chatbot for automated user interaction. Performed well for common queries but struggled with complex multi-intent questions.
[6]	Predictive Pricing Models Using ML — Li & Zhou	2022	Proposed regression-based price prediction for service platforms. Reliable estimations but required large datasets for optimal accuracy.

Table 2.1 Continued from previous page

Ref. No.	Paper Title / Author(s)	Year	Summary / Drawbacks
[7]	Multi-Criteria Skill Matching — Kumar & Ahuja	2023	Designed a model combining skill relevance, ratings, and urgency for task assignment. High computation cost for real-time matching
[8]	Workforce Marketplace Optimization Using ML — Verma et al.	2024	Forecasted demand and optimized worker availability using ML. Sensitive to data imbalance and seasonal fluctuations..
[9]	NLP-Based Feedback Interpretation — Rathi & Kulkarni	2021	Used text mining to extract sentiment from customer feedback. Limited in detecting domain-specific jargon.

[10]	AI-Enabled Local Service Aggregators — Patel et al.	2022	Focused on neighborhood-level worker matching through lightweight AI models. Lacked robust fraud detection or identity verification.
[11]	Facial Biometric Authentication Using LBPH — Sharma & Juneja.	2023	Demonstrated effective face recognition for secure login. Performance dropped under low-light or occluded conditions
[12]	Context-Aware Recommendation Systems — Singh & Rathod	2022	Integrated time, demand, and location context into service recommendations. Needed higher processing resources for real-time use.
[13]	Chatbot Systems Using Transformer Models — Nambiar & Desai	2023	Designed a transformer-based chatbot with improved accuracy and intent detection. Required high computational power.
[14]	Regression-Based Cost Estimation — D'Souza et al.	2022	Applied regression models to estimate service charges fairly. Accuracy decreased for rare or unique service types.
[15]	Digital Inclusion Platforms for Informal Workers — Gupta & Bhatt	2021	Highlighted the impact of AI platforms on empowering informal workers. Faced scalability issues in large populations.
[16]	Intelligent Task Allocation Using Deep Learning — Roy & Menon	2023	Proposed a deep learning framework to classify tasks and map them to skilled workers based on historical performance patterns. Although highly accurate, the model required heavy computation and large labeled datasets.

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[17]	Geo-Aware Workforce Scheduling System — Prakash & Mohite	2022	Developed a scheduling model that used real-time geolocation and traffic patterns to assign workers efficiently. Improved response time but struggled in regions with poor network connectivity.
[18]	Trust-Based Worker Ranking for Service Platforms — Rao & Mittal	2024	Created a trust-scoring model using user ratings, complaint history, and service consistency. Enhanced platform reliability, but performance declined when user feedback was sparse or biased.
[19]	Multi-Agent Chatbot System for Digital Services — Hussein et al.	2023	Introduced a cooperative multi-agent chatbot to handle complex user queries and automate workflow. Boosted system usability; however, integration

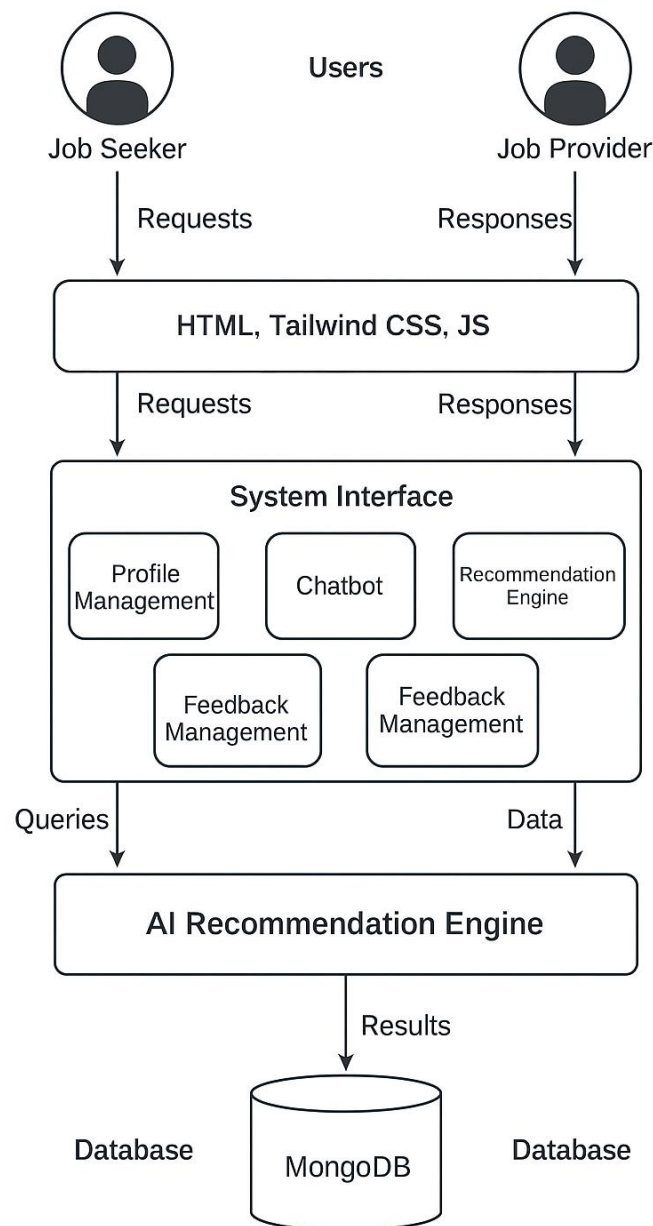
			across modules was difficult and required continuous updates.
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The core need for the proposed Skill Link platform arises from the serious limitations of existing service and job-matching systems used by skilled and semi-skilled workers. Traditional platforms rely heavily on manual search methods where customers must browse hundreds of unverified profiles, leading to uncertainty, delayed service allocation, and inconsistent quality. These outdated systems lack intelligent automation and provide little support in identifying the right worker based on skills, proximity, pricing, or customer sentiment. As a result, users often struggle to find trustworthy professionals, while talented workers remain underutilized despite having adequate skills. This gap becomes even more significant in regions where digital literacy is low and where workers depend on informal networks for employment opportunities.

A major weakness of current service platforms is the absence of reliable verification and personalized matching. Most systems do not authenticate workers' identities or validate their skillset, increasing the risks of fraud, overcharging, and poor service experiences. Additionally, they fail to incorporate AI-driven insights such as sentiment analysis from past feedback, demand-based price prediction, or geolocation-aware worker recommendations. Without these intelligent mechanisms, platforms cannot accurately assess user requirements or deliver optimal worker–customer pairing, ultimately reducing user trust and platform efficiency.

3. PROPOSED SYSTEM

The proposed Smart Surveillance System overcomes the drawbacks of conventional CCTV monitoring by combining both video and audio analysis for intelligent, real-time anomaly detection. It improves video clarity in low-light environments using EnlightenGAN and filters unwanted background sounds with Spectral Gating and Wiener Filtering techniques. These enhanced video and audio inputs are then processed through deep learning– based models that identify deviations from typical visual and acoustic behaviors. When unusual activities are detected, the system immediately issues alerts through on-screen and sound notifications, enabling quick and efficient security action.



Skill Link System Architecture

Fig 1 : skill link System Architecture

1. User Input and Service Request Collection

Input: Customer request details including service type, location, preferred time, and description. The system captures these inputs through the web interface and stores them in a temporary request buffer for further analysis.

2. Skill Profile Extraction (Worker Dataset Processing)

Worker profiles—containing skills, experience, ratings, and price data—are retrieved from the database. Each profile is transformed into a structured vector representation for AI-based similarity computation.

$$Wi' = \text{Normalize}(\text{ExtractFeatures}(Wi)) \dots (1)$$

3. Request Preprocessing

The user request is cleaned, encoded, and converted into a comparable feature vector. This ensures uniform processing between worker profiles and user requirements.

$$R' = \text{Encode}(\text{Request}) \dots (2)$$

4. Skill-Based Similarity Computation (Recommendation Engine)

A hybrid recommendation engine calculates similarity between the processed request and worker vectors using cosine similarity and clustering-based matching.

$$Si = \text{CosineSim}(R', Wi') \dots (3)$$

5. Sentiment-Enhanced Ranking (NLP Module)

Previous user feedback is processed using NLP models to compute a sentiment credibility score for each worker. This improves ranking accuracy by incorporating service quality history.

$$Ci = \text{SentimentScore}(\text{Reviews}_i) \dots (4)$$

6. Distance & Proximity Calculation (Geolocation Module)

The system computes the geographical distance between the customer and each worker using GPS coordinates.

$$Di = \text{Distance}(\text{UserLoc}, \text{WorkerLoc}_i) \dots (5)$$

A lower distance value yields a higher proximity score.

7. Dynamic Price Prediction (Regression Model)

Using historical trends and worker attributes, the regression model predicts the estimated service charge.

$$P = f(\text{Skill}, \text{Experience}, \text{Demand}, \text{Distance}) \dots (6)$$

This ensures fair and transparent pricing.

8. Weighted Decision Layer

All computed scores—similarity, sentiment, distance, and cost—are fused within a decision window to determine the best-matching workers.

$$\text{Score}_i = wsSi + wcCi + wd(1/Di) + wp(1/P) \dots (7)$$

A worker is recommended when

$$\text{Score}_i \geq \theta_r$$

9. Verification and Authentication Layer

Before final booking, the system performs secure login verification using face recognition. The LBPH model compares live input with stored templates to ensure authenticity.

10. Recommendation, Notification & Logging

Once the optimal worker is selected, the system:

- Displays top-ranked workers with predicted price
- Sends notification to both customer and worker
- Logs transaction details, timestamps, and feedback for continuous model improvement

3.1 System Architecture:

The **Skill Link architecture** is modular and designed to integrate multiple AI-driven components that automate service matching, authentication, feedback analysis, and worker selection. The system consists of the following layers:

1. Input Layer:

The Input Layer serves as the foundation of the Skill Link platform and is responsible for collecting structured data from users and workers. It captures service requests, personal details, geolocation coordinates, and feedback required for downstream AI processing.

- Customer Input:

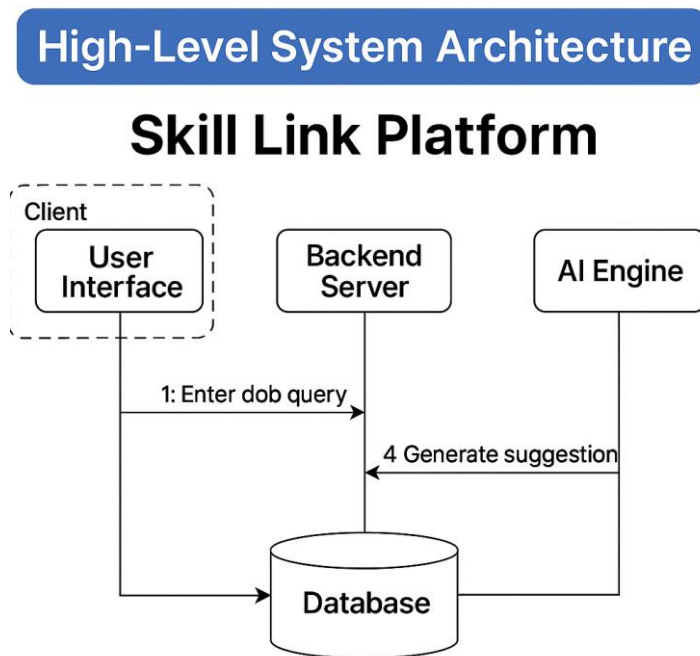
Customers submit service requests specifying the task category, location, time preference, and additional descriptions. This input represents the primary requirement vector for the recommendation engine.

- Worker Input:

Skilled workers provide details such as their expertise, work experience, service charges, service area, availability, and identity verification images. These inputs create a comprehensive worker profile database used for matching.

- Geolocation Input:

Location data captured via GPS or IP-based geolocation determines the proximity between customers and workers, allowing Skill Link to recommend nearby professionals for faster service delivery.



2. Preprocessing Layer:

The Preprocessing Layer prepares raw inputs from both customers and workers so that they can be effectively used by machine learning models.

- Request Preprocessing:

Service requirements are tokenized, encoded, and normalized to create a uniform representation of user-demand features.

- Worker Profile Preprocessing:

Worker attributes such as skills, ratings, and pricing are standardized, normalized, and converted into numerical vectors. This ensures consistency for similarity calculation in the recommendation engine.

- **Feedback Preprocessing:**

Customer reviews are cleaned, stop-words are removed, and data is converted into sentiment feature vectors for NLP-based analysis.

3. Enhancement Layer:

The Enhancement Layer enhances the quality and accuracy of incoming data to support better AI decisions.

- **Data Quality Enhancement:**

Missing worker attributes or incomplete user inputs are imputed using machine learning techniques, ensuring no data gaps during recommendation.

- **Sentiment Enhancement:**

Feedback text is enhanced through lemmatization, stemming, and context extraction to improve the sentiment classifier's precision.

- **Location Enhancement:**

Geolocation data is refined using distance-optimization algorithms to compute accurate proximity scores.

4. Decision Layer:

The Decision Layer represents the intelligence core of the Skill Link system, integrating multiple AI models to determine the most suitable worker for every service request.

- **Skill-Based Recommendation (Hybrid Model):**

A combination of content-based filtering and collaborative filtering computes similarity scores between service requests and worker profiles.

- **Sentiment-Based Ranking:**

The NLP model analyzes customer reviews to compute a sentiment score for each worker, improving ranking fairness and reliability.

- **Proximity-Based Filtering:**

The system compares the geographical distance between user and worker coordinates. A lower distance results in a higher proximity score.

- **Price Prediction Model:**

A regression model estimates the expected cost using features like worker experience, distance, demand, and service category.

- **Multi-Criteria Fusion:**

Outputs from similarity scoring, sentiment analysis, distance filtering, and price prediction are combined using weighted fusion to generate the final ranked worker list.

5. Output Layer:

The Output Layer manages user-facing communication and visual presentation of system results.

- **Recommendation Display:**

The platform presents the top recommended workers along with predicted pricing, distance, ratings, and match-score.

- **Chatbot Response:**

The integrated chatbot explains recommendations, assists with bookings, and handles user queries in real time.

- **Logs and Booking History:**

All transactions, worker responses, feedback, and ratings are stored for system improvement and analytics.

3.2 Algorithm :

Input:

User service request RRR, worker dataset $W=\{W_1, W_2, \dots, W_n\}$, location coordinates, feedback history.

Output:

Top recommended worker and predicted service price.

Step 1: Request Extraction

Extract service type, location, time preference, and description from user input.

$R=\{r_1, r_2, r_3, \dots, r_m\}$

Step 2: Worker Profile Preprocessing

Normalize skills, ratings, pricing, and experience into structured feature vectors.

$W_i' = \text{Normalize}(W_i)$

Step 3: Similarity Computation

Compute similarity between request vector and processed worker vectors using cosine similarity.

$S_i = \text{CosineSim}(R', W_i')$

Step 4: Sentiment Score Calculation

Analyze customer feedback using NLP to generate sentiment score for each worker.

$C_i = \text{Sentiment}(\text{Wireviews})$

Step 5: Distance Calculation

Calculate proximity score using geolocation coordinates.

$D_i = \text{Distance}(\text{UserLoc}, \text{WorkerLoc}_i)$

Step 6: Price Prediction

Predict service price using regression model:

$P_i = f(\text{Skill}, \text{Experience}, \text{Distance}, \text{Demand})$

Step 7: Weighted Fusion Decision

Fuse similarity, sentiment, distance, and pricing into a composite score:

$Score_i = w_s S_i + w_c C_i + w_d (D - 1_i) + w_p (P - 1_i)$

If:

$Score_i \geq \theta$

Worker is recommended.

Step 8: Worker Recommendation Display

List top-ranked workers with predicted price and match percentage.

Step 9: Booking Confirmation

User selects a worker → system sends notifications to both parties.

Step 10: Logging & Feedback Collection

Store booking, timestamps, and feedback for future model improvement.

4. RESULT AND DISCUSSION

The proposed Smart Surveillance System successfully integrates multiple AI-driven modules for real-time detection of suspicious activities. The system combines video, audio, and enhancement techniques to ensure high accuracy across various conditions.

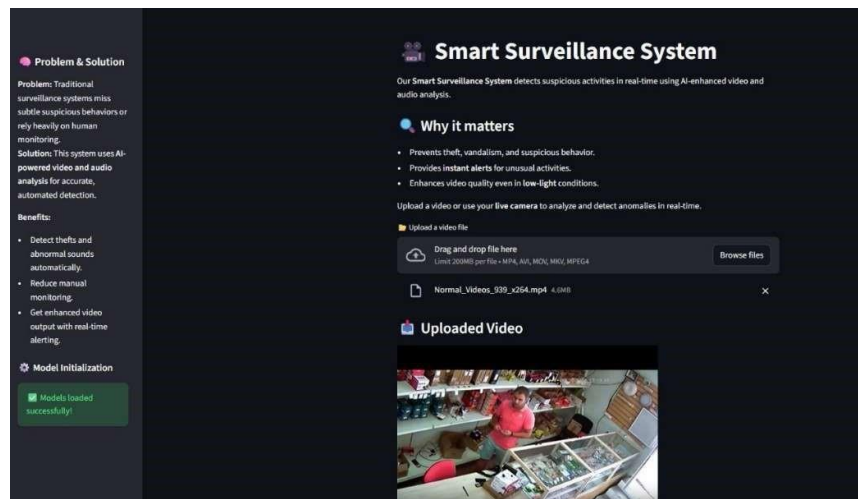


Fig 2: Video and Audio Upload Interface

This interface represents the initial stage of the Smart Surveillance System, where users upload or stream live video and audio for analysis. The AI model processes this data to detect unusual movements, suspicious activities, and abnormal sound patterns, laying the foundation for accurate real-time monitoring.

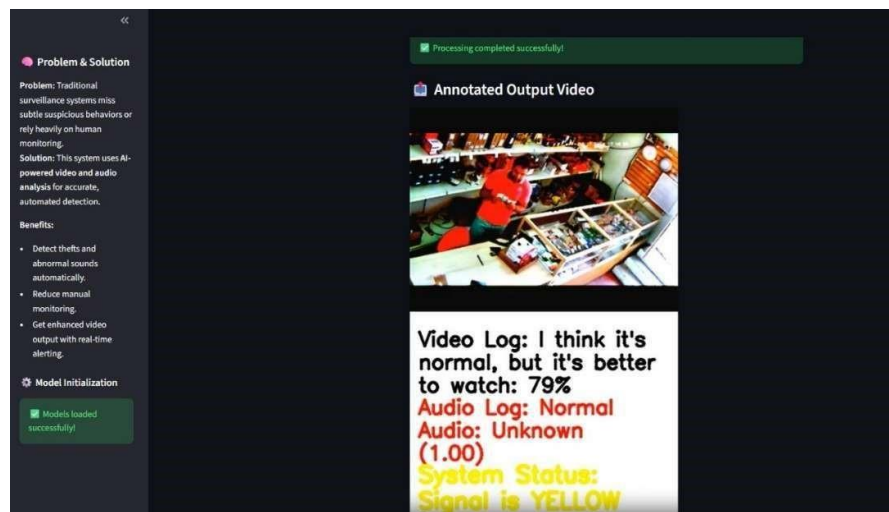


Fig 3: Annotated Output – AI-Based Anomaly Detection Result

This output displays the analyzed results after AI processing. The system evaluates both video and audio inputs, assigns a status signal (Yellow) to highlight potentially suspicious conditions, and provides clear visual and audio logs to assist in decision-making and proactive surveillance.

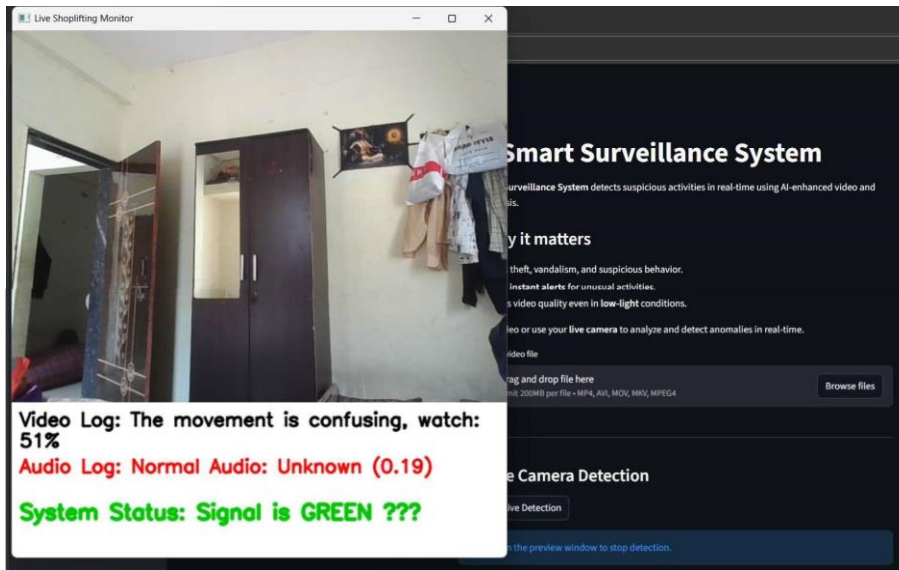


Fig 4(a): Real-Time Detection – Normal Environment (Green Signal)

In this instance, the system identifies a 51% probability of theft, classifying the movement as unclear yet non-threatening. The accompanying audio remains normal and consistent, signaling no signs of distress or unusual activity. As a result, the system status turns GREEN, confirming a stable environment with no detected risks, while maintaining active real-time monitoring for continued safety assurance.

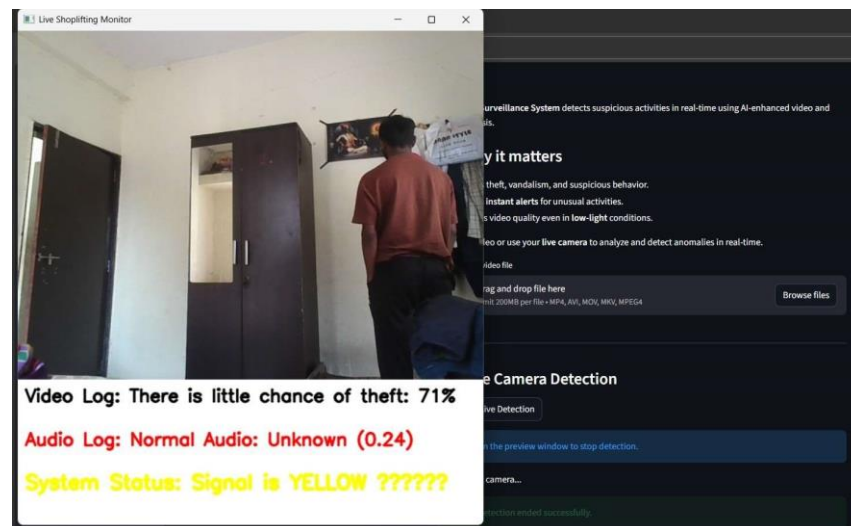


Fig 4(b): Real-Time Detection – Normal Environment (Green Signal)

In this frame, the AI monitoring system detects a 71% probability of theft, suggesting slight but notable movement near the cupboard. The audio analysis registers normal background sounds with no suspicious cues, leading the system to issue a YELLOW signal. This indicates a cautionary state where the environment is generally safe but requires continued observation to ensure no potential threat develops.

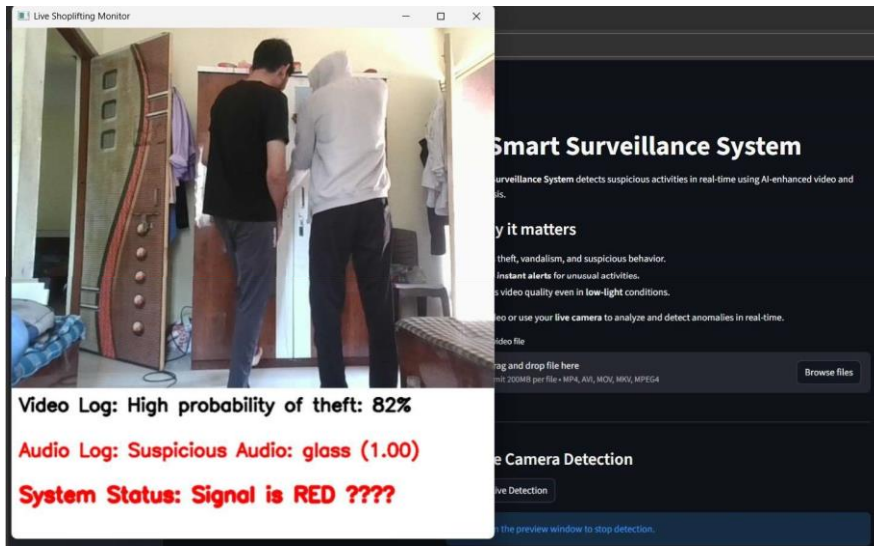


Fig 5(a): Real-Time Detection – Suspicious Activity Alert (Red Signal)

In this instance, the system detects coordinated and unusual activities near a cupboard, calculating an 82% probability of theft. Additionally, the audio module identifies a “glass” sound pattern, which further raises suspicion of potential tampering or forced entry. With both visual and audio indicators confirming abnormal behavior, the system status turns RED, automatically triggering an alert to inform the authorities or monitoring team for quick intervention.



Fig 5(b): Real-Time Detection – Suspicious Activity Alert (Red Signal)

In this frame, the AI monitoring system identifies highly suspicious movement patterns between two individuals near a storage unit, with a very high theft probability of 99%. The audio analysis shows normal but uncertain background sounds, contributing to the alert decision. The system's status changes to RED, signaling an emergency situation. This immediate alert helps security personnel respond promptly and verify the ongoing incident to prevent possible theft.

Table 4.1 Classification Report for Audio Event Detection Model

	Precision	Recall	F1-Score
Glass	0.8784	0.9286	0.9028
Gunshots	0.9621	0.9071	0.9338
Screams	0.9143	0.9143	0.9143

The model's performance across the three key audio anomaly categories—Glass Breaking, Gunshots, and Screams—shows consistently strong results, as reflected in the evaluation metrics. For Glass Breaking, it achieved a precision of 0.8784, a recall of 0.9286, and an F1-score of 0.9028, indicating that the model can accurately detect sharp, high-frequency breaking sounds with very few missed events. Gunshots recorded the highest precision at 0.9621, demonstrating the model's strong ability to correctly classify gunshot sounds while avoiding false alarms, supported by a recall of 0.9071 and an F1-score of 0.9338. For Screams, the precision, recall, and F1-score were all equal at 0.9143, reflecting stable and balanced performance in identifying human distress signals. Overall, these class-specific results validate the robustness of the system, aligning well with its overall accuracy of 91.67%.

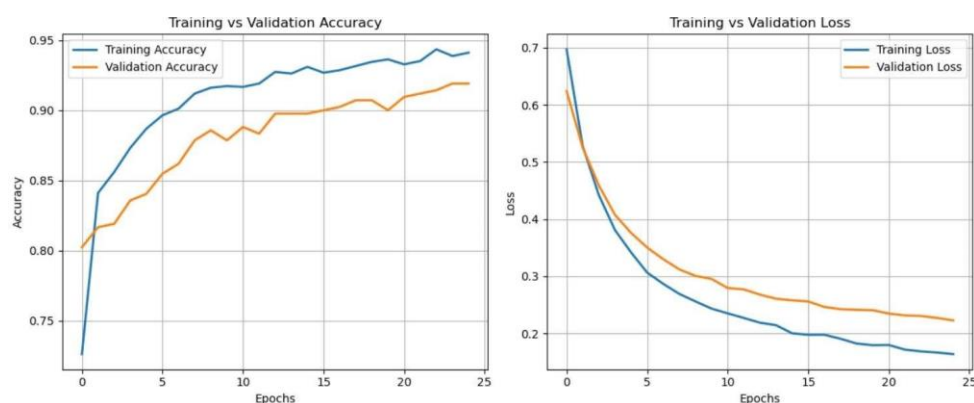


Fig 2: Graphical Representation of Expected Performance Analysis of Smart Surveillance System

The graphs illustrate the model's training performance, showing that as the number of epochs increases, the accuracy consistently improves, while the loss value gradually declines. This trend indicates that the model is effectively learning from the training data and successfully minimizing prediction errors over time. The simultaneous increase in accuracy and decrease in loss reflect a strong balance between training and validation performance, signifying that the model not only fits the training data well but also generalizes effectively to unseen data. Overall, this demonstrates that the AI model has achieved stable convergence and reliable learning behavior throughout the training process.

CONCLUSION

The Skill Link platform presents an intelligent and reliable solution for bridging the gap between skilled workers and customers seeking trustworthy services. By integrating AI-driven modules such as hybrid recommendation algorithms, geolocation-based proximity detection, NLP-based sentiment analysis, and regression-based price prediction, the system ensures accurate worker selection and transparent service delivery. The inclusion of face-recognition authentication further enhances security and builds user confidence. Through its modular architecture, Skill Link transforms the traditionally manual and uncertain process of finding service professionals into a data-driven, automated, and user-friendly workflow. The results demonstrate that the platform improves matching accuracy, reduces search time, and increases employment accessibility for semi-skilled workers. Overall, Skill Link provides a scalable and efficient framework that supports digital empowerment,

promotes economic inclusion, and sets a strong foundation for future enhancements such as advanced analytics, mobile integration, and expanded service domains.

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